Assignment: Implement K-Nearest Neighbors (KNN) Algorithm

Implement Python code to build a **KNN algorithm from scratch** based on the pseudocode provided in the image. Follow the instructions step by step

Input:

1. D: a set of training samples $\{(\vec{x}_1, y_1), \ldots, (\vec{x}_n, y_n)\}$
2. k: the number of nearest neighbors
3. $d(\vec{x}, \vec{x}')$: a distance metric
4. $\vec{x}$: a test sample
5. **for each** training sample $(\vec{x}_i, y_i) \in D$ **do**
6. Compute $d(\vec{x}, \vec{x}_i)$, the distance between $\vec{x}$ and $\vec{x}_i$
7. Let $N \subseteq D$ be the set of k training samples with the smallest distances
8. Compute the label:

~~5~~ **return** $\hat{y}$

1. Training Data

Let D be a set of training samples $\{(\vec{x}_1, y_1)$,……., $\{(\vec{x}_n, y_n)\}$

- The training samples are shown in the table below.
- In our dataset, $n = 20$, each sample has 3 features, and the labels are binary (0 or 1).
- In the KNN algorithm, the training data is stored in memory rather than being used to build an explicit model.
- Different data storage strategies may be used to optimize access speed, depending on the implementation.

No.	Feature 1	Feature 2	Feature 3	Label
1	21.8	3.4	279	0
2	29.9	5.8	240	0
3	34.4	8.1	285	0
4	30.3	8.9	251	1
5	36.9	8.6	197	0
6	31.1	4.1	290	0

7	21	3.4	268	0
8	22.1	8.4	256	0
9	25.9	4.6	287	1
10	38.2	6.5	230	1
11	28.1	9.4	208	1
12	37.2	4.5	289	1
13	19.6	3.8	299	1
14	33.1	4.6	262	1
15	28.3	4.2	161	0
16	21.8	5.8	189	1
17	19.8	5.7	259	0
18	24.9	4.6	264	1
19	36.1	9.5	259	0
20	32.1	4.3	288	0

Dataset reference: https://www.kaggle.com/datasets/chik0di/health-and-lifestyle-dataset?select=health_lifestyle_dataset.csv

2. Input Parameter

- **k**: Number of nearest neighbors to consider.
- Input **k** from the user, ensuring:
  0<k<=n, n is number of training samples

3. Distance Calculation

Based on the input features are three$x'(x_1, x_2, x_3)$, the distance can be calculated using the following equation:

d(x, x') = $\sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + (x_3 - x'_3)^2}$

4. Assignment value to x': test samples

We want to predict $x'(20, 3.4, 161)$

5. Iterate over the training samples to calculate distances.

Perform iteration on all 20 training samples to compute the distance matrix, and then proceed to calculate the weight matrix.

6. Calculate distance

Since we have 20 training data, so after calculate distances the output is matrix align with our data labels

No.	Feature 1	Feature 2	Feature 3	Label	Distance(d(x,q)
1	21.8	3.4	279	0	$\sqrt{(x_1-q_1)^2+(x_2-q_2)^2+(x_3-q_3)^2}$
2	29.9	5.8	240	0	
3	34.4	8.1	285	0	
4	30.3	8.9	251	1	
5	36.9	8.6	197	0	
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20	32.1	4.3	288	0	

7. Chose the k nearest neighbors

Use Numpy to find the k smallest distances among the n training samples.

- Calculate weight using this formular $weight = \frac{1}{d}$ or $weight = \frac{1}{1+d}$.
  Using in $weight = \frac{1}{1+d}$ to prevent errors when d=0.
  If k =3, then the only the three smallest distance are used to compute the weights.

8. Compute the Label:

Using this formular

$$\hat{y} = \arg\max_{c} \sum_{i \in N_k} w_i \cdot \mathbf{1}(y_i = c)$$